



- 1 (a) State what is meant by a vector quantity.

.....
 [1]

- (b) A sphere falls vertically through a liquid that has density 830 kg m^{-3} . The sphere has radius r and constant velocity v , as shown in Fig. 1.1.

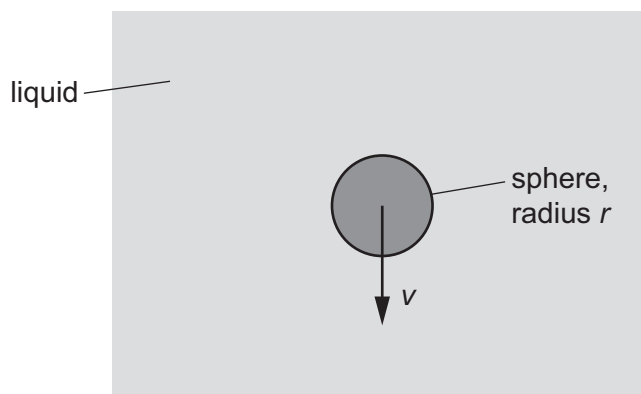


Fig. 1.1

- (i) The drag force D acting on the sphere is given by

$$D = 6\pi r\eta v$$

where η is a property of the liquid.

Determine the SI base units of η .

SI base units [3]

- (ii) State an equation showing the relationship between the magnitudes of the weight W , drag force D and upthrust U acting on the sphere.

..... [1]





(iii) The volume of the sphere is 4.6 cm^3 . The drag force D is 0.32 N .

Calculate the weight of the sphere.

weight = N [2]

[Total: 7]

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